

Stakeholder-Specific Architecture Viewpoints for EU Regulation: A Generative AI-Supported Method

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European Union regulation imposes organisational obligations whose implications for individual stakeholders are often difficult to identify. Translating these obligations into representations of daily activities, responsibilities, and dependencies requires a systematic connection between legal requirements and operational evidence. This dissertation proposes a human-supervised method, supported by Generative Artificial Intelligence (GenAI), for identifying, specifying, and materialising stakeholder-specific architecture viewpoints.

Following Design Science Research Methodology (DSRM) and grounded in ISO/IEC /IEEE 42010, the method combines stakeholder interviews, operational footprint extraction, regulatory relevance analysis, viewpoint specification, and ArchiMate model generation. Traceability links architectural elements and relationships to binding legal text and documented operational evidence, while explicit ownership distinctions, human approval gates, and cross-artifact reconciliation constrain generated content.

The method is demonstrated using the Digital Operational Resilience Act (DORA) legislation and two stakeholders in a banking context. The resulting Architecture Description (AD)s relate regulatory requirements to role-specific work while distinguishing direct responsibilities, external dependencies, and uncertain ownership. Preliminary stakeholder feedback indicates that the formal views can support regulatory understanding and reveal missing operational context, enabling viewpoint refinement. It also identifies limitations in explanatory narratives and simplified diagrams. The demonstrations support the feasibility of the approach, with opportunities for further development. The contribution is a traceable process for communicating stakeholder-specific regulatory impact without assessing organisational compliance.

Additional Key Words and Phrases: European Union, Conceptual Modelling, DORA Regulation, ArchiMate, Architecture Description, Generative AI

1 Introduction

European Union legislation, including the Digital Operational Resilience Act, imposes obligations that affect organisational governance, operational processes, and technical systems. Although these obligations are generally assigned to regulated entities or legally defined roles, their implementation depends on activities and decisions distributed across individual stakeholders.

The problem addressed in this dissertation is how to communicate the implications of those obligations for a particular stakeholder's work. Legal requirements and compliance matrices can identify duties without making their relationship to operational responsibilities and dependencies explicit. This work proposes a human-supervised method that leverages Generative Artificial Intelligence (GenAI) to connect legal requirements with documented operational evidence and expresses the resulting relationships through stakeholder-specific architecture viewpoints and views.

1.1 Context and Motivation

The legal framework is a major constraint for enterprises, particularly in regulated sectors facing an increasing volume and complexity of legislation [Blanco-Lainé et al. 2019]. Regulations such as Digital Operational Resilience Act (DORA) express requirements for

governance, organisational processes, and information technology, but their practical effect is operational. Conventional compliance checklists and requirement matrices can record legal duties, yet often provide limited visibility into how those duties affect the responsibilities, dependencies, and concerns of a particular stakeholder. Enterprise Architecture (EA) and the ArchiMate modelling language provide a basis for addressing this communication problem through viewpoints that select and relate the architectural information relevant to a stakeholder's concerns.

Building on the previous thesis on the conceptual modelling of Electronic Identification, Authentication and Trust Services (eIDAS2) and Network and Information Systems (NIS2), this work proposes a human-supervised method for identifying, specifying, and materialising stakeholder-specific ArchiMate viewpoints in the DORA context. The method connects binding legal requirements with documented operational evidence and uses GenAI as a controlled accelerator and method stressor to support creation and review of Architecture Descriptions (ADs). The resulting artefacts are intended to make stakeholder-relevant regulatory impacts, responsibilities, and dependencies available for inspection and feedback, without providing authoritative legal interpretation, automating compliance decisions, or replacing legal and architectural judgement.

1.2 Objectives

The primary objective of this research is to develop a method, aligned with ISO/IEC/IEEE 42010, for identifying and formalising stakeholder-specific viewpoints that translate legislative requirements into ArchiMate representations of relevant operational footprints. Following Design Science Research Methodology (DSRM), the research designs and demonstrates a purposeful artefact: a human-supervised, GenAI-supported methodology for eliciting, specifying, and materialising regulation-aware viewpoints [Peffer et al. 2007].

DSRM structures the research through problem identification, objective definition, artefact design and development, demonstration, and evaluation. The artefact developed through this research process is the twelve-stage methodology presented later. Its application to DORA provides a demonstration, while stakeholder feedback and examination of the generated artefacts provide qualitative evidence on the value of the artefact.

Its secondary objective is to produce ADs that intended stakeholders can understand, inspect, challenge, and enrich. Stakeholder feedback may identify missing operational context, corrections, concerns, or scope refinements that can improve subsequent viewpoint generations.

The research proposition is that a human-supervised method can systematically derive traceable stakeholder-specific ArchiMate viewpoints by connecting binding legal provisions with documented

operational evidence under explicit scope, relationship, and approval rules. These viewpoints make relevant regulatory impacts, responsibilities, dependencies, and operational context available for structured inspection without automating legal interpretation or compliance decisions.

2 Conceptual Modelling and Enterprise Architecture

Conceptual modelling is about describing the semantics of software applications at a high level of abstraction. Specifically, conceptual modellers:

- Describe structure models in terms of entities, relationships, and constraints;
- Describe behaviour or functional models in terms of states, transitions among states, and actions performed in states and transitions;
- Describe interactions and user interfaces in terms of messages sent and received and information exchanged.

Although conceptual modelling is commonly applied to individual software applications, the same principles also support the understanding and communication of enterprise contexts. Enterprise conceptual models represent relevant business, organisational, information, and technical concepts at an appropriate level of abstraction. Anaby-Tavor et al. found that business analysts, business architects, and solution consultants used such modelling-related work products to understand business contexts and inform requirements or business recommendations [Anaby-Tavor et al. 2010]. At this scale, conceptual modelling identifies and expresses relevant concepts, relationships, and constraints, while EA provides an integrated perspective on organisational structures, processes, information systems, and technical infrastructure [Lankhorst et al. 2004].

An enterprise model should therefore select the concepts and relationships relevant to a particular concern rather than reproduce every organisational detail. This selection is particularly important where obligations, risks, or changes cross organisational and technical boundaries. ISO/IEC/IEEE 42010 provides a foundation for expressing enterprise architectures through structured architecture descriptions without equating a representation with the enterprise itself [ISO/IEC/IEEE 2022]. In this work, ArchiMate provides the consistent EA notation used to express the selected concepts and relationships [ope 2022].

Selecting relevant enterprise information requires an explicit account of whose concerns are being addressed and how those concerns should be represented. 3 provides the concepts used to organise that selection.

3 Architecture Descriptions, Viewpoints and Views

An AD is a work product used to express an architecture. It may take the form of a document, a collection of models, a repository, or another medium, and organises information concerning stakeholders, concerns, viewpoints, views, and model kinds. Within EA, such models contribute to ADs, which ISO/IEC/IEEE 42010 defines as work products used to express an architecture rather than the architecture itself. An AD may take the form of a document, a collection of models, a repository, or another medium, and brings together

elements such as stakeholders, concerns, viewpoints, views, and model kinds [Bolinhas 2025; ISO/IEC/IEEE 2022].

ISO/IEC/IEEE 42010 formalises the distinction between viewpoints and views. A viewpoint defines the conventions, modelling methods, notations, and analytical techniques used to address a particular set of stakeholder concerns. A view is the corresponding part of an AD that expresses the architecture of an Entity of Interest (EoI) according to those conventions [ISO/IEC/IEEE 2022]. A viewpoint may address governance, operational, technical, or logical concerns, and may govern multiple views for different stakeholders or levels of detail [ope 2022; The Open Group 2022].

For example, a viewpoint addressing controlled change can specify which stakeholder concerns, operational concepts, legal requirements, and relationship types must be considered. A corresponding Controlled Change view then represents the particular testing, approval, deployment, and recording activities evidenced for a stakeholder. The viewpoint establishes the modelling conventions; the view applies them to a specific operational context.

An Architecture Description Language (ADL) provides the concepts, relationships, and notation needed to express ADs consistently. In this work, ArchiMate is used as the EA modelling language because it provides a standard set of elements, relationships, and iconography for representing integrated enterprise architectures [ope 2022]. It therefore supports the consistent expression of stakeholder-specific regulatory requirements, operational responsibilities, and dependencies.

4 European Normative Acts

Every action taken by the European Union (EU) is based on treaties. They are the starting point for EU law and are hence referred to as primary law.

Currently there are five types of juristic acts, including regulations, directives, decisions, recommendations and opinions¹.

- Regulations - Legal acts that are binding in their entirety. They are directly and uniformly applicable to all EU member states as soon as they enter into force, without needing to be transposed into national law.
- Directives - Set binding objectives upon EU member states to achieve a certain result, but leave them free to choose how to achieve these objectives.
- Decisions - Binding in their entirety, can specifies those to whom it is addressed, being binding only on them.
- Recommendations - Non-binding. They allow the EU institutions to make their views known and to suggest a line of action without imposing any legal obligation.
- Opinions - Also non-binding, opinions allow EU institutions to make a statement without imposing any legal obligation.

4.1 Regulation as an Architectural Constraint

Regulation functions as a major architectural driver. As noted by Bolinhas[Bolinhas 2025], “the legal framework is a significant constraint for all businesses, notably when dealing with an increasing number of regulations and directives” [Blanco-Lainé et al. 2019].

¹Types of EU law can be found here: https://commission.europa.eu/law/law-making-process/types-eu-law_en

These constraints may affect different dimensions. Regulations like DORA mandate the creation of specific organizational units, or roles, such as the Lead Overseer. This effectively “draws” the organizational map, implementing a structural constraint. These constraints also exist in a behavioural form, since legislations describe rules, which are fundamentally rights and obligations [Breaux and Antón 2008] describing what can and cannot be done. Other constraints, like data protection laws, impose constraints on how to handle business objects, defining how data can be stored, for how long and who has access rights to the data. DORA also imposes a type of “supply chain” dependencies, requiring financial organizations to audit and constrain their third-party service providers [DOR 2022].

Organisations may be subject to several interacting legal requirements, whose implementation can constrain architectural choices. ADs can make these constraints and dependencies explicit and support comparison between implementation alternatives within the applicable legal boundaries. Resolving conflicts between regulations is outside the scope of the demonstrations presented here, which examine stakeholder-specific representations of DORA.

4.2 Legislation Structure

Identifying the structure of an EU legal act is necessary for locating the provisions represented by architectural requirements. The Interinstitutional Style Guide describes the principal components of an act²: [Publications Office of the European Union 2021]

- (1) **Title**;
- (2) **Preamble**(Citations, Recitals, Solemn Forms);
- (3) **Enacting Terms** (and Annexes).

The title already holds a lot of information regarding the act, such as title, type, date of adoption and reference number.

The preamble is next, and provides the background, purpose and legal basis of the law before the legally binding rules begin.

Lastly, the enacting terms are the part of the act that sets the new rules. Enacting terms are composed of numbered articles that can be grouped into sections, chapters, titles and parts (in this order, listed from the greatest granularity level to the least). Any of these groupings can contain articles, which may also be supplemented by annexes, where necessary. A good overview the European legislation can be seen in figure 1.

5 Related work

The following subsections approach Artificial Intelligence (AI) in the context of this work. It is important to note that this work builds on builds on the value created by a previous dissertation, “Conceptual Modelling of Legal Compliance Requirements - The cases of eIDAS2 and NIS2”. This provided a methodical and structured approach, composed of four main steps: The creation of an Excel Traceability Matrix as an external legal reference; Developing an ArchiMate Index for internal architectural navigation; Connecting both through a hierarchical numbering system, and Constructing the ADs according to the defined architectural principles.

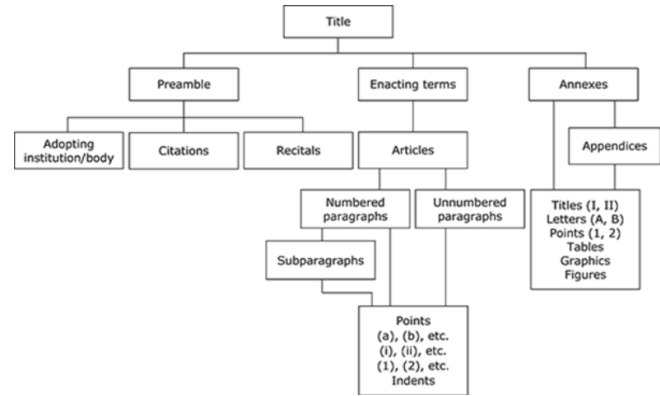


Fig. 1. Structure of an act [Publications Office of the European Union 2023]

5.1 AI in Regulatory Analysis and Modelling

Requirements engineering has progressively moved from manual methods towards automated support intended to reduce effort, time, and human error. Early automated approaches focused primarily on requirements analysis, using Natural Language Processing (NLP) techniques such as part-of-speech tagging and syntax parsing to generate Unified Modelling Language (UML) models and perform consistency checks [Umar and Lano 2024]. However, such approaches struggled with the ambiguity and contextual complexity of real-world legal documents [Zadenoori et al. 2025].

Generative Large Language Model (LLM)s extend this automation to requirements elicitation, validation, and modelling by synthesising content rather than only detecting or classifying defects [Zadenoori et al. 2025]. Their outputs may nevertheless be unreliable when relevant context is unavailable. Retrieval-Augmented Generation (RAG) addresses this limitation by retrieving information from enterprise knowledge bases or legal sources to ground model responses [Jeong 2023].

In regulatory Requirements Engineering (RE), MUlti-layered Regulatory Change Identification and Analysis (MURCIA) uses language models to assist human analysts in identifying and interpreting regulatory changes across textual, semantic, deontic, and pragmatic layers, with an evaluation reporting F1 scores above 90% for textual and semantic change identification [Abualhaija et al. 2024]. Conversational modelling approaches also allow users to describe and iteratively refine model syntax through natural-language interaction [Härer 2023]. Nevertheless, current work remains insufficiently integrated with complex industrial and multi-regulatory workflows, leaving a need for systematic methods that connect regulatory analysis with formal EA representations [Elahidoost 2024; Zadenoori et al. 2025].

5.2 Multi-Agent Debate

Multi-agent debate (MAD) is a collaborative reasoning approach in which multiple LLM agents analyse the same problem, exchange and challenge arguments, and produce a collective conclusion. By making disagreement explicit, it aims to improve accuracy, reduce individual bias, strengthen factual grounding, and provide clearer justification for the final output [Gómez et al. 2026; Oriol et al.

²Structure and a guide on how to read EU law can be found on: <https://fabianbohnberger.com/2024/04/10/how-to-read-eu-law/comment-page-1/>

2025]. Its design involves the selection of participants and roles, the interaction mechanism through which agents exchange information, and an agreement mechanism such as voting, scoring, or a judge-based decision [Oriol et al. 2025].

Independent initial assessments can preserve genuine differences of interpretation before agents are influenced by competing arguments. Later rounds allow agents to rebut, refine, or withdraw their positions; however, additional rounds also increase execution time, token consumption, and cost, often with only marginal gains [Oriol et al. 2025]. Structured approaches organise this process into initialisation, argumentation, adjustment, and synthesis, with a moderator controlling turn-taking, preserving debate history, and resolving or recording remaining conflicts [Gómez et al. 2026].

This work applies MAD to regulatory-relevance assessment. Differentiated evaluators analyse legal relevance from literal, systemic, and gap-oriented perspectives, while a moderator records and challenges their reasoning. Explicit consensus is required for inclusion, whereas unresolved cases are preserved for human review, supporting conservative and traceable relevance decisions.

6 Problem Analysis

A normative act may affect many organisational units, systems, processes, and governance mechanisms, making a regulation-wide architectural representation too broad for an individual stakeholder. Stakeholders therefore require views that foreground the responsibilities and dependencies relevant to their own work, rather than a universal representation containing substantial irrelevant detail [Breaux et al. 2008; Kosenkov et al. 2025]. ISO/IEC/IEEE 42010 and ArchiMate address this need through viewpoints and views, which define and present architectural information for specified stakeholders and concerns [Ope 2022; ISO/IEC/IEEE 2022]. The resulting artefacts must consequently be comprehensible to their intended stakeholders (**R1**).

This requires bridging a legal-to-operational relevance gap. European legislation, including DORA, generally assigns duties to regulated entities, management bodies, or other legally defined roles rather than to individual employees [DOR 2022]. Legal requirements must therefore be related to a stakeholder’s documented responsibilities, activities, systems, dependencies, and concerns. This is an interpretative RE activity because legal and operational sources use different levels of abstraction, terminology, and perspectives [Ghanavati et al. 2014].

GenAI can support this work by structuring legal and operational information, identifying candidate relationships, and producing material for review [Abualhaija et al. 2024]. It cannot, however, serve as legal, operational, or architectural evidence: hallucinations and non-deterministic outputs may produce unsupported conclusions [Autio et al. 2024]. The methodology therefore requires human approval (**R2**), stable results (**R3**), traceability to legal sources (**R4**), to operational evidence (**R5**), and prohibits unsupported inference (**R6**). These requirements provide criteria for examining the method and its outputs, and are summarized in 1. The demonstrations provide preliminary evidence against these criteria. Human reviewers

inspect, reject, and approve material conclusions, preserving accountability while using GenAI as a controlled analytical aid rather than an autonomous compliance assessor.

7 Methodology

The methodology goes through 12 stages, ranging from the initial prompt optimization to the delivery of a fully informative model. These are identified as the corresponding number in the following subsections. Multiple agents work throughout every step, making use of a plethora of skills that provide the necessary knowledge basis and restrict LLM behaviour. Throughout all the stages, the LLM questions the user when it feels it does not have all the necessary information (to prevent hallucination, bolstering **R6**), and before any decision is locked in, ensuring human approval throughout the process (**R2**).

To understand the human interactions with the methodology, it is crucial to understand the roles that interact with it. The user is the person who runs the methodology and approves progression between stages while the stakeholder supplies operational knowledge and comments on whether the resulting representations reflect their work.

7.1 Traceability Matrix Evolution

The Excel Traceability Matrix evolved from a manual process to a fully automated one. A program was created to analyse the HyperText Markup Language (HTML) from the target legislation, and output the Traceability Matrix without any human input.

Besides automation for the sake of efficiency, the traceability matrix also suffered a complete redesign to improve readability (**R1**). The baseline Traceability Matrix used an Identifier (ID) system that was not intuitive. This is because the different IDs can refer to any of the sub-article levels, which are not explained anywhere in the matrix. Additionally, of all the groupings referred in section 4.2, only Chapters are mentioned. To address this, every possible grouping that a legislative text may have exists as an ID column (**R4**). An excerpt of the new traceability matrix can be observed in table 2.

7.2 Context Discovery

The twelve phases of the methodology are organised into three groups. The first, Context Discovery, comprises stages one to five and establishes the evidence base required for later viewpoint scoping and materialisation.

1 - The first stage, Prompt Optimization, converts the initial user request into a bounded and unambiguous pipeline input. It preserves the intended objective, introduces explicit constraints, and records missing information as placeholders rather than supplying assumptions. The approved prompt is retained as an artefact, ensuring that the initial scope and subsequent decisions remain visible throughout the methodology (**R6**).

2 - Interview Design then produces a jargon-free and regulation-blind interview guide for eliciting the stakeholder’s operational footprint (**R1**). Questions address the stakeholder’s mission, workflows, assets, dependencies, resilience practices, and governance

Table 1. Assessment Criteria for the Methodology Requirements

Requirement	Proposed assessment criterion
R1 – Comprehensibility	Stakeholders can explain represented responsibilities and dependencies and identify omissions or inaccuracies.
R2 – Human approval	Consequential inclusion and modelling decisions have recorded human approval.
R3 – Stability	Repeated executions with unchanged inputs produce materially consistent relevance, ownership, and relationship decisions.
R4 – Legal traceability	Each represented legal duty resolves to an identifiable provision in the selected legal source.
R5 – Operational traceability	Each represented operational claim or relationship is supported by documented stakeholder evidence.
R6 – Evidence discipline	Unsupported claims remain excluded or explicitly unresolved instead of being incorporated as established facts.

Table 2. Sample from the traceability matrix of DORA generated by the traceability matrix creator program

Part	Title	Chapter	Section	Article	Paragraph	Sub-Paragraph	Point	Point Sub-Paragraph	Sub-Point	Indent	Text
		I									General provisions
		I		1							Subject matter
		I		1	1						In order to achieve a high common level of digital operational resi
		I		1	1		a				requirements applicable to financial entities in relation to:
		I		1	1		a		i		information and communication technology (ICT) risk managemer
		I		1	1		a		ii		reporting of major ICT-related incidents and notifying, on a volunt
		I		1	1		a		iii		reporting of major operational or security payment-related incide
		I		1	1		a		iv		digital operational resilience testing;
		I		1	1		a		v		information and intelligence sharing in relation to cyber threats ar
		I		1	1		a		vi		measures for the sound management of ICT third-party risk;
		I		1	1		b				requirements in relation to the contractual arrangements conclud
		I		1	1		c				rules for the establishment and conduct of the Oversight Framewo
		I		1	1		d				rules on cooperation among competent authorities, and rules on s
		I		1	2						In relation to financial entities identified as essential or important
		I		1	3						This Regulation is without prejudice to the responsibility of Memb

context, supported by alternative angles and fallbacks. The user conducts the interview and records the responses, producing the documented operational evidence required by the following stages (R5).

3 - The third stage, Operational-Footprint Extraction, analyses the interview responses and records distinct operational facts using OP-* identifiers. Explicit relationships are separately captured as REL-OP-* records, including their endpoints, direction, and supporting interview evidence (R5 and R6). Tasks, assets, dependencies, approvals, incident interfaces, and friction points are reviewed and approved separately.(R2)

4 - Stakeholder and Concern Identification uses the approved operational footprint to determine the stakeholder's organisational position and identify every documented concern, bottleneck, friction point, or operational gap. Each concern receives a CON-* identifier and, where available, retains its originating OP-* evidence (R5).

5 - Regulatory Relevance Analysis is the fifth stage and evaluates the approved operational footprint, stakeholder classification, and concerns against the selected EU legislation. Three evaluators independently assess the legal text from literal, systemic, and gap-oriented perspectives, retrieving and analysing the regulation article by article. Each candidate article must be justified through an operational anchor, a legal bridge, and an ownership assessment (R5 and R6). A moderated debate then takes place to attempt to resolve disagreements. Only unanimously accepted articles proceed as relevant, while unresolved cases remain visible for user review, supporting stable relevance decisions (R3).

For each approved article, the process records atomic legal duties as REQ-* requirements and resolves them against verified canonical

legal-text identifiers in the generated traceability matrix (R4). A Coverage Ledger then connects requirements to documented OP-* or CON-* records through COV-* entries. The resulting REL-COV-* relationships record the operational and legal endpoints, supporting evidence, effect, and ownership state, thus preserving traceability to operational evidence and preventing unsupported relationships (R5, R6).

Finally, the process maps the stakeholder through the employing organisation to the applicable law-defined entity or role group through evidence-backed REL-ROLE-* relationships (R5). This distinction prevents an individual employee from being incorrectly represented as the regulated entity. The resulting report is approved through gates covering article selection, omitted candidates, legal anchors, coverage relationships, and legal-role placement, becoming the approved evidence base for the next methodological group (R2).

7.3 Viewpoint Definition

The second group of the methodology comprises stages six to eight and converts the approved evidence base into a formal stakeholder viewpoint. It begins with a technology-selection gate, in which the user selects the technology used to materialise the immutable regulation overview baseline (R2).

The regulation overview and the stakeholder-specific views have different scopes. The overview is derived from the complete binding text within the selected legal corpus and preserves the regulation-wide legal context. Stakeholder-specific views are subsequently derived from the narrower set of duties and operational relationships justified by the relevance and coverage analysis (R5 and R6).

6 - Three technology choices are available. Direct ArchiMate Extensible Markup Language (XML) generation produces a native .archimate model that can be opened in Archi; LikeC4 provides an experimental architecture-as-code alternative based on a concise Domain Specific Language (DSL), although its primary scope is software rather than enterprise architecture; pyArchimate generates an ArchiMate model through a Python script. Regardless of the technology selected, the overview retains the same legal core (**R3**): Binding duties are organised into requirement families, anchored to verified legal text (**R4**), connected to the 'stakeholder' - 'organisation' - 'legal-role' chain, validated for connectivity, and approved before later stages may use it.

7 - Viewpoint Scope determines which approved legal and operational material should be represented for a stakeholder, and why. Ownership remains explicit, distinguishing work directly performed by the stakeholder from externally owned, constrained, or unclear controls (**R6**). The user approves the boundary, included requirements, streams, and connectivity of the proposed views (**R2**).

8 - Viewpoint Generation converts the approved scope into a formal ArchiMate viewpoint specification. For each proposed view, it defines its purpose, stakeholder perspective, concerns, operational streams, permitted ArchiMate concepts and relationships, requirement-family structure, and visual rules (**R1**). The regulation overview remains unchanged as the regulation-wide legal baseline (**R3**), while the additional views represent the narrower stakeholder-specific scope.

This stage also produces `7_view_graph.json`, an authoritative machine-readable contract for later narrative, slide, and model generation. It records every permitted node, approved REL-* relationship, evidence reference, ownership safeguard, requirement-family nesting rule, and stakeholder anchor for each view (**R5**). The user validates both the ArchiMate metamodel rules derived from the approved scope and the complete view graph before the viewpoint specification and graph are persisted, supporting stable and evidence-traceable later artefacts (**R3**, **R4** and **R5**).

7.4 View Generation

The final methodological group comprises stages nine to twelve and materialises artefacts that communicate how the selected legislative act affects the operational footprint of the stakeholder. The produced artifacts are a textual explanation (phase 9), markdown diagrams (phase 10), and ArchiMate views (phase 11).

The approved view graph is the reference against which generated artefacts are checked. It records the permitted elements, relationships, ownership states, and evidence references. Conformance to this graph establishes consistency with the approved specification (**R3**).

9 - Textual View Explanation produces a per-view narrative from the approved viewpoint and view graph. It explains the regulatory and operational context of each view and how its elements and relationships should be interpreted (**R1**). The resulting explanation also includes a traceability table connecting coverage entries to their operational evidence, legal requirements, ownership states, and representing elements (**R4** and **R5**). The user approves each view name and description (**R2**).

10 - Markdown Slide Diagrams provide a condensed, presentation-oriented representation of the overview and each stakeholder-specific view. Each slide contains an ASCII sketch of the principal operational path, dependencies, and requirement families, supported by tables listing granular requirements and approved relationships. The slide is therefore meant to be an accessible summary rather than a complete graphical projection of the underlying view graph, and is accompanied by a plain-language operational takeaway (**R1**). Each slide and its supporting information require user approval before persistence (**R2**).

11 - View Generation then materialises the approved viewpoint as a native architecture model that can be opened, inspected, reused, and refined. The selected technology (direct ArchiMate XML, LikeC4, or pyArchimate) receives the regulation overview, viewpoint specification, authoritative view graph, explanation, and slides. The view graph remains the source of truth (**R3**), while the narrative and slides support the organisation and presentation of the native model.

The generator creates all approved model elements and relationships while preserving requirement-family containment, stakeholder anchors, ownership safeguards, and legal documentation (**R4**). A stakeholder is represented as realising a requirement only when the relevant work is directly performed. Otherwise, dependencies, constraints, approvals, or external roles are represented instead. The output is validated against the selected technology and reconciled with the view graph: missing or additional elements, altered relationships, broken containment, disconnected views, or changes to the overview cause rejection (**R3** and **R6**). Proposed layouts are presented to the user for approval before the model is saved (**R2**).

12 - Lastly, Cross-Artifact Reconciliation verifies that the overview, viewpoint and graph, textual explanation, slides, native model, and legal traceability matrix communicate the same approved information (**R3**). Using the graph as the ground truth, it checks legal anchors, operational chains, view membership, relationships, ownership semantics, containment, and technology-specific model validity (**R4**, **R5** and **R6**). Any discrepancy is documented in an audit report and corrected only after user approval, after which the audit is repeated (**R3**). This ensures that the different artefacts can support different readers without silently communicating incompatible regulatory or operational claims.

8 Results and Demonstration

The method was demonstrated with two banking stakeholders: a back-end developer and an infrastructure and cloud engineer with database-administration responsibilities. These cases illustrate different operational contexts within the same regulatory domain, covering software delivery and technical platform administration. Feedback was collected during presentations of the generated views and explanatory artefacts to the stakeholders. The observations provide qualitative evidence of usefulness and limitations within these cases.

8.1 Operational Context

Both cases began with an optimised prompt and a stakeholder interview used to establish the operational footprint and analysis boundary. In the back-end developer case, the prompt retained placeholders to delimit the available context and objectives, whereas the Infrastructure and Cloud Engineer prompt was considered sufficiently complete without them. Both were used to produce an interview to be conducted with the designated stakeholder by the user. The infrastructure interview was sent to the stakeholder instead of being conducted, resulting in the feedback that some questions were vague. Nevertheless, the responses provided sufficiently detailed information.

For the back-end developer, the analysis boundary concerns work on an internal Data Platform. The stakeholder develops varied services as well as framework changes, code tests, creates pull requests, coordinates releases and monitors production deployments. Their work depends on functional specifications, permissions, management and solution-architecture decisions, infrastructure deployment support, cloud-native infrastructure and core platform services. The resulting footprint contains 89 approved, interview-grounded records. The stakeholder accepted the extracted footprint, classification, and concerns without amendment.

For the Infrastructure and Cloud Database Administrator, the boundary concerns the administration of distributed query engines, metadata management, and high-performance databases within the bank's internal data-technology layer. The stakeholder provisions access, maintains technologies through updates and configuration changes, validates deployments, investigates connection and usage problems, and supports incident response. Their work depends on infrastructure, networking, technology-support, security, and authorisation teams, as well as enterprise platforms for access governance, release automation, security management, observability, and Information Technologies (IT) service operations. This case produced 58 approved operational records, whose footprint and concerns the stakeholder considered accurate.

Both boundaries distinguish technical work directly performed by the stakeholder from decisions and controls owned elsewhere. The developer implements and tests changes but lacks final authority for architecture, functional requirements, and production approval. The infrastructure stakeholder remains constrained by DevOps-change, access, and production-authorisation processes. Both participate in incident handling within defined escalation paths, although the infrastructure stakeholder also restarts, investigates, and mitigates database outages before producing a post-incident report. These distinctions provide the evidence base for subsequently determining which regulatory effects can be attributed to the stakeholder, represented as external constraints, or excluded from the viewpoint.

8.2 DORA Relevance and Ownership

In both cases, regulatory relevance analysis compared the approved operational footprint with DORA through the multi-agent debate panel. Rather than treating every entity-level duty as a personal stakeholder responsibility, the analysis retained only requirements supported by an operational chain and recorded their legal anchors and ownership states. This enables the resulting views to distinguish

controls directly performed by the stakeholder from constraints, external dependencies, and responsibilities for which the available evidence does not establish an owner.

In the back-end developer case, the selected DORA scope concentrated on duties affecting platform development, controlled change, testing, monitoring, continuity, and incident-related work. Duties concerning the wider Information and Communication Technology (ICT) risk-management framework, crisis communication, incident classification, and reporting to competent authorities were excluded because the interview did not evidence developer participation. A deliberate manual addition of Article 28 was used to test intervention in the relevance process. The resulting cloud-risk material remained weak because the evidence did not establish a direct effect on the developer's work, demonstrating that the method retains the distinction between legal applicability to the bank and relevance to an individual stakeholder.

For the Infrastructure and Cloud Engineer case, the panel selected a broader set of requirements spanning the ICT risk-management lifecycle, including platform administration, access and security, controlled change, monitoring, recovery, incident handling, and resilience testing. This wider scope reflected the stakeholder's documented responsibility for configuring and maintaining core data query, governance, and storage platforms, as well as supporting access management, testing, incident response, and service recovery. In contrast, management-body governance, authority reporting, and entity-level third-party-risk management remained outside the stakeholder boundary because they were not evidenced as part of the role.

The two cases therefore produced different relevance boundaries while applying the same evidence and ownership rules. The developer viewpoint emphasised the delivery lifecycle and its dependencies on decisions and controls owned by other teams, whereas the Infrastructure and Cloud Engineer viewpoint covered a broader technical-administration and resilience context. In both cases, pyArchimate was selected to produce the overview model, but the distinct operational evidence resulted in different requirement-family structures. The resulting views preserve the wider DORA legal context while limiting direct stakeholder responsibility to what the operational evidence supports.

8.3 Resulting Architecture Description

Both stakeholder cases produced pyArchimate ADs from approved viewpoints, preserving the regulation overview, legal requirements, operational evidence, ownership states, and ArchiMate relationships. In each case, the architecture description separates work directly performed by the stakeholder from external organisational controls, dependencies, and areas where ownership could not be established. The resulting models are therefore intended as traceable communication and refinement artefacts rather than as assessments of organisational compliance.

For the back-end developer, the resulting AD contains the regulation overview and six stakeholder-specific thematic views: ICT Asset Governance, Protection and Change, Detection and Continuity, Incident Learning, Resilience Testing, and Microsoft Cloud Risk. This structure reflects the developer's delivery-oriented operational

These outcomes should be interpreted within the scope of the demonstrations. GenAI supported the extraction, organisation, and

representation of information, but human review, approval gates, traceability, and cross-artifact reconciliation remained necessary to constrain its use. Moreover, the completeness of a viewpoint depends on the information made explicit during elicitation: omitted activities or controls may remain unrepresented or have unclear ownership. The generated views can therefore serve not only as communication artefacts, but also as feedback mechanisms through which stakeholders identify omissions and enrich subsequent iterations. Broader assessment with additional stakeholders and independent reviewers remains necessary.

Future work should improve the presentation and explanation of generated views. Although the current models are structurally valid, their layout still requires manual refinement to improve readability, reduce relationship crossings, and better reflect stakeholder reading paths. Automated layout and connector-routing strategies could reduce this effort. Textual explanations should also move beyond describing the methodology itself and instead explain the regulatory and operational significance of the view from the perspective of the stakeholder.

The methodology could also be extended in its legal and technical foundations. Incorporating Regulatory Technical Standards (RTS) and Implementing Technical Standards (ITS) would enable the method to represent more detailed obligations concerning areas such as ICT change management, testing, source-code testing, and logging. Further work is also needed to improve traceability-matrix parsing and identifier schemes for amendment-heavy legislation, as well as to update the pipeline for ArchiMate 4.

Beyond stakeholder communication, the methodology could support enterprise gap detection and version-aware regulatory analysis. Comparing viewpoints obtained from multiple employees, particularly across equivalent roles, could reveal recurring indications of organisational gaps that warrant review, although a single interview cannot establish non-compliance. Recording regulatory versions and effective dates could additionally enable the method to identify which requirements, viewpoints, and models are affected by later legal amendments, allowing targeted maintenance rather than repetition of the complete pipeline.

References

2022. *The Open Group Standard ArchiMate® 3.2 Specification*. Retrieved December 9, 2025 from <https://pubs.opengroup.org/architecture/archimate32-doc/singlepage/>
- European Union 2022. *REGULATION (EU) 2022/2554 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 14 December 2022 on digital operational resilience for the financial sector and amending Regulations (EC) No 1060/2009, (EU) No 648/2012, (EU) No 600/2014, (EU) No 909/2014 and (EU) 2016/1011*. European Union. <https://eur-lex.europa.eu/eli/reg/2022/2554/oj/eng>
- Sallam Abualhaija, Marcello Ceci, Nicolas Sannier, Domenico Bianculli, Lionel C. Briand, Dirk Zetsche, and Marco Bodellini. 2024. AI-enabled Regulatory Change Analysis of Legal Requirements. In *2024 IEEE 32nd International Requirements Engineering Conference (RE)*. IEEE.
- Ateret Anaby-Tavor, David Amid, Amit Fisher, Avivit Bercovici, Harold Ossher, Matthew Callery, Michael Desmond, Sophia Krasikov, and Ian Simmonds. 2010. Insights into Enterprise Conceptual Modeling. *Data & Knowledge Engineering* 69, 12 (2010), 1302–1318. doi:10.1016/j.datak.2010.10.003
- Chloe Autio, Reva Schwartz, Jesse Duniety, Shomik Jain, Martin Stanley, Elham Tabassi, Patrick Hall, and Kamie Roberts. 2024. *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. Technical Report NIST AI 600-1. National Institute of Standards and Technology. doi:10.6028/NIST.AI.600-1
- Gaëlle Blanco-Lainé, Jean-Sébastien Sottet, and Sophie Dupuy-Chessa. 2019. *Using an Enterprise Architecture Model for GDPR Compliance Principles*. Springer International Publishing.
- Paulo Alexandre Pereira Bolinhas. 2025. *Conceptual Modelling of Legal Compliance Requirements - The cases of eIDAS2 and NIS2*. Master's thesis. Instituto Superior Técnico.
- Travis D. Breaux and Annie I. Antón. 2008. Analyzing Regulatory Rules for Privacy and Security Requirements. *IEEE Transactions on Software Engineering* (Jan. 2008). doi:10.1109/TSE.2007.70746
- Travis D. Breaux, Annie I. Antón, Kent Boucher, and Merlin Dorfman. 2008. Legal Requirements, Compliance and Practice: An Industry Case Study in Accessibility. In *Proceedings of the 16th IEEE International Requirements Engineering Conference (RE 2008)*. IEEE, 43–52. doi:10.1109/RE.2008.36
- Parisa Elahidoost. 2024. Towards a Tool Supported Approach for Regulatory Requirements Engineering. In *2024 IEEE 32nd International Requirements Engineering Conference (RE)*. IEEE, Reykjavik, Iceland. doi:10.1109/RE59067.2024.00067
- Sepideh Ghanavati, André Rifaut, and Eric Dubois. 2014. Goal-Oriented Compliance with Multiple Regulations. In *2014 IEEE 22nd International Requirements Engineering Conference (RE)*. IEEE.
- Susana Gómez, Alejandro Mozo, Tomás Navarro, Sergio Gálvez, and Francisco L. Valverde. 2026. Multi-Agent Debate System Based on Large Language Models: Structured Deliberation and Validation in Satellite Communications. Research Square preprint. Posted May 11, 2026. doi:10.21203/rs.3.rs-9575030/v1
- Felix Härer. 2023. Conceptual model interpreter for Large Language Models. In *ER 2023: Companion Proceedings of the 42nd International Conference on Conceptual Modeling*. CEUR Workshop Proceedings.
- ISO/IEC/IEEE. 2022. *ISO/IEC/IEEE 42010:2022 Software, systems and enterprise — Architecture description*. Second edition.
- Cheonsu Jeong. 2023. A Study on the Implementation of Generative AI Services Using an Enterprise Data-Based LLM Application Architecture. *arXiv preprint arXiv:2309.01105v2* (2023). <https://arxiv.org/abs/2309.01105>
- Oleksandr Kosenkov, Michael Unterkalmsteiner, Daniel Mendez, and Jannik Fischbach. 2025. Regulatory Requirements Engineering in Large Enterprises: An Interview Study on the European Accessibility Act. In *Product-Focused Software Process Improvement: 25th International Conference, PROFES 2024 (Lecture Notes in Computer Science, Vol. 15452)*, Dietmar Pfahl, Javier Gonzalez Huerta, Jil Klünder, and Hina Anwar (Eds.). Springer, 204–220. doi:10.1007/978-3-031-78386-9_14
- Marc M. Lankhorst, René van Buuren, Diederik van Leeuwen, Henk Jonkers, and Hugo ter Doest. 2004. Enterprise Architecture Modelling—The Issue of Integration. *Advanced Engineering Informatics* 18, 4 (2004), 205–216. doi:10.1016/j.aei.2005.01.005
- Marc Oriol, Quim Motger, Jordi Marco, and Xavier Franch. 2025. Multi-Agent Debate Strategies to Enhance Requirements Engineering with Large Language Models. *arXiv:2507.05981 [cs.SE]* <https://arxiv.org/abs/2507.05981>
- Ken Peffers, Tuure Tuunanen, Marcus A. Rothenberger, and Samir Chatterjee. 2007. A Design Science Research Methodology for Information Systems Research. *Journal of Management Information Systems* 24, 3 (2007), 45–77. doi:10.2753/MIS0742-1222240302
- Publications Office of the European Union. 2021. *2. Structure of a Legal Act*. <https://style-guide.europa.eu/en/content/-/isg/topic?identifier=2-structure-legal-act>
- Publications Office of the European Union. 2023. *Interinstitutional Style Guide*. European Union, Luxembourg. Section 2: Structure of an Act. Accessed: September 19, 2026. <https://www.scribd.com/document/624373174/EU-Interinstitutional-Style-Guide>
- The Open Group. 2022. *The TOGAF Standard, 10th Edition — Introduction and Core Concepts*. The Open Group. <https://pubs.opengroup.org/togaf-standard/introduction-and-core-concepts/chap04.html>
- Muhammad Aminu Umar and Kevin Lano. 2024. Advances in automated support for requirements engineering: a systematic literature review. *Requirements Engineering* 29 (2024), 177–207.
- Mohammad Amin Zadenoori, Jacek Dąbrowski, Waad Alhoshan, Liping Zhao, and Alessio Ferrari. 2025. Large Language Models (LLMs) for Requirements Engineering (RE): A Systematic Literature Review. *arXiv preprint arXiv:2509.11446v1* (2025).

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